

Silverline House — 25-Storey Residential HRB

Structural Calculation Pack

Part A — Structure | BS EN 1992 / 1993 / 1997

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1 Introduction and Scope

This document forms part of the Gateway 2 application for Building Control Approval submitted to the Building Safety Regulator under the Building Safety Act 2022. The information presented herein has been prepared in accordance with the CLC Guidance Suite for new Higher-Risk Buildings and the requirements set out in Guidance Note 04 (Application Information Schedule) and Guidance Note 06 (Document Management and Submission). All design work described in this document has been carried out by suitably qualified engineers and specialists with relevant experience in higher-risk residential buildings.

The scope of this document covers the primary structural system, including the reinforced concrete frame, post-tensioned flat slab floors, core walls, and the ground-bearing and piled foundation system. The document does not cover secondary structures such as facade support brackets (addressed separately in the Facade Interface and Support Design document), internal partition framing, or fit-out works. All dimensions and coordinates are in millimetres unless otherwise stated. Grid references correspond to the structural grid shown on drawing SLH-STR-001 Rev A.

2 Design Standards and Reference Documents

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

All reinforced concrete elements are specified in C32/40 concrete (characteristic cylinder / cube strength) with a maximum water-cement ratio of 0.55 and minimum cement content of 300 kg/m³. Post-tensioned slab tendons are specified in low-relaxation 15.7 mm strand to BS 5896, stressed to 75% of characteristic tensile strength. Reinforcement throughout is Grade B500B to BS 4449. Concrete cover to reinforcement is 35 mm to internal elements and 50 mm to external elements exposed to the weather, in accordance with BS EN 1992-1-1 Table 4.4N for exposure class XC3. Minimum cement content and maximum water-cement ratio comply with the durability requirements of BS 8500-1 for the intended service life of 50 years.

3 Structural Scheme Description

A site investigation was carried out in accordance with BS EN 1997-2 and BS 5930. The investigation comprised twelve rotary core boreholes to a maximum depth of 30 m below existing ground level, together with in-situ standard penetration tests at 1.5 m intervals. Laboratory testing was carried out on selected samples to determine soil classification, shear strength parameters, consolidation characteristics, and chemical contamination. The ground conditions comprise approximately 3 m of made ground overlying Terrace Gravels (3–9 m depth), London Clay (9–22 m), and Lambeth Group deposits below 22 m. Groundwater was encountered at 2.5 m depth in the gravels. The design groundwater level for structural purposes is taken as 0.5 m below existing ground level.

The foundation system consists of 600 mm diameter continuous flight auger (CFA) piles installed to a minimum depth of 18 m below existing ground level, penetrating at least 3 m into the Lambeth Group. Pile design has been carried out in accordance with BS EN 1997-1 using a combination of ground model analysis and empirical correlations from the site investigation data. The working load per pile is 2,800 kN in compression and 800 kN in tension. Settlement analysis indicates maximum pile head settlements of 8 mm under working loads, with differential settlement between adjacent piles not exceeding 5 mm. Pile load testing in accordance with BS EN ISO 22477-1 is required during construction, with at least two maintained load tests to 150% of the working load prior to commencing pile cap construction.

4 Structural Analysis — Gravity Loads

Design loads have been derived in accordance with BS EN 1991-1-1 and BS EN 1991-1-3 for imposed, snow and wind loading respectively, with the UK National Annex applied throughout. The imposed load for residential floors is 1.5 kN/m² with an additional 1.0 kN/m² for partitions. Plant rooms at roof level are designed for 5.0 kN/m². Wind loading has been calculated for a basic wind speed of 23 m/s (10-minute mean at 10 m height in open country) adjusted for the site location and topography in accordance with BS EN 1991-1-4 and the UK National Annex. The net lateral wind force on the building is 3,200 kN at the 50-year return period, resisted by the reinforced concrete core.

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4.1 Lateral Stability Analysis

Lateral stability is provided by a reinforced concrete core of plan dimensions 12 m × 8 m positioned at the centre of the building footprint. The core walls are 300 mm thick below Level 5, reducing to 250 mm above Level 5 and 200 mm above Level 15. Core wall reinforcement has been designed for combined bending, shear and axial forces using the General Method of BS EN 1992-1-1. The inter-storey drift under the 50-year wind load is limited to $H/500 = 17$ mm to satisfy serviceability requirements. The building has been checked for progressive collapse in accordance with BS EN 1991-1-7 Annex A, adopting the notional member removal procedure for all floors above ground level.

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4.2 Façade Loading

Facade loading has been assessed in accordance with BS EN 1991-1-1 for permanent actions and BS EN 1991-1-4 for wind actions on the facade panels. Dead load contributions from the facade system are applied as line loads to the slab edge at each floor level.

The cladding system is assumed to impose a dead load of 2.5 kN/m^2 on the slab edge at each floor level. This value has been derived from the cladding contractor's preliminary design data and includes the self-weight of panels, rails, brackets, insulation, and all fixings. This load has been applied in the structural model as a uniformly distributed line load along the perimeter slab edge beams.

Wind pressures on the facade are applied as normal forces to the cladding panels and transferred to the bracket array. The design wind pressure at the top of the building is 1.05 kN/m^2 (positive pressure) and -1.35 kN/m^2 (suction), derived from BS EN 1991-1-4 with the UK National Annex. The facade bracket design is contained in the Facade Interface and Support Design document (ref 2.3.3).

5 Foundation Design Summary

The foundation system consists of 600 mm diameter continuous flight auger (CFA) piles installed to a minimum depth of 18 m below existing ground level, penetrating at least 3 m into the Lambeth Group. Pile design has been carried out in accordance with BS EN 1997-1 using a combination of ground model analysis and empirical correlations from the site investigation data. The working load per pile is 2,800 kN in compression and 800 kN in tension. Settlement analysis indicates maximum pile head settlements of 8 mm under working loads, with differential settlement between adjacent piles not exceeding 5 mm. Pile load testing in accordance with BS EN ISO 22477-1 is required during construction, with at least two maintained load tests to 150% of the working load prior to commencing pile cap construction.

A post-tensioned transfer slab at Level 1 spans between the core walls and the perimeter columns at 7.5 m centres, supporting eight residential floors above. The slab is 600 mm deep with unbonded tendons in both directions at 200 mm centres. The design has been analysed using finite element software with validation by hand calculation for critical sections. Shear checks at column heads comply with BS EN 1992-1-1 clause 6.4 with punching shear reinforcement provided at all columns. The transfer slab is classified as a Critical Structural Element for inspection purposes; the inspection and maintenance strategy is defined in the Fire and Emergency File.